20 Question Quiz w/ Answers: Test Your Web Handling Knowledge? c

Web Handlers Quiz (with Answers)

Question 1: Tension Control

How do you determine the target web handling tension for a give product in units of force or force per width?

Test your web in a tensile elongation tester per ASTM D638 or ISO 527. Find the force per width tension that damages your product, either yielding (for ductile materials) or breaking (for brittle materials). Set you average tension to 10-20% of the level that damages your product.

If you know you have modest machine or crossweb direction tension variations from roller variations, web bagginess (especially if you web is narrow), misaligned splices, and roller drag / inertia, then you can run with less safety factor, if needed.

More on this topic was covered in this archived Web Lines column:

What Is the Right Tension? (Dec. 2009)

Question 2: Tension Control

Name three causes of tension variations across the web's width (i.e. things would create a loose edge or center)?

- Roller variations, including: misalignment (level or tram), deflection, diameter variations vs. width.
- Crossweb dimensional variations, either permanent (bagginess or skew) or temporary (from temperature or moisture variations)
- Splicing misalignments

Bonus Question: How do you know which you have?

- Roller variations will create a local bagginess that will not appear throughout the process or correspond to the splice as it passes through the process.
- Permanent web bagginess and skew will be seen throughout the web lines, possibly changing in magnitude and location as a roll unwinds and commonly changing from input roll to input roll. Temporary dimensional variations from temperature or moisture will occur where the process or environment induced these changes.
- Splice-related bagginess will be temporary as the splice passes through the equipment and quickly disappear after the splice goes by.

Be warned: these tension variations causes can occur simultaneously.

More on this topic was covered in this archived Web Lines column:

What Is the Right Tension? (Dec. 2009)

Question 3: Tension Control

Name one advantage of using a dancer roller has over a tension load cell roller in a closed-loop tension control system?

- The dancer will accumulate and dispense web when the input web speed differs from the output web speed for a limited amount of time. In this regard, a dancer roller can be a mechanical system that forgives electrical and controls problems.
- A tension load cell roller has almost no ability to accumulate or dispense web.

More on this topic was covered in this archived Web Lines column:

Dancer Rollers: Trust But Verify Nov. 2006

Question 4: Tension Control

Name one advantage of using a tension load cell roller has over a dancer roller in a closed-loop tension control system?

- A tension load cell roller, when properly calibrated, provides a signal proportional to the web tension.
- A dancer roller provides some feedback on input to output speed variations, but does not indicate tension variations.
- Load cell rollers can be used with limited wrap angle and areas where the web can only be contacted on one side.
- Load cell rollers will tend to hold their alignment better.
- Holding their alignment combined with their potentially smaller wrap angle means less likelihood of wrinkles relative to dancers.

More on this topic was covered in this archived Web Lines column:

Well Done Tension Sept. 2002

Question 5: Tension Control

Name one advantage and one disadvantage of draw control (a.k.a. as speed ratio control)?

Advantages:

- Draw control is the simplest intermediate tensioning systems, requiring no rollers or feedback between driven rollers, allowing it to fit where driven rollers must fit into a tight space.
- Draw control lets you set a percent stretch of processing your material independent of tension, which can be important to products where elongation is a better predictor of damage than tension.
- Draw control is used in machine direction registered processes. The speed adjustments of closed-loop tension or torque control would create shifts in printing and die-cutting registration.

Disadvantages:

- Draw control is widely misunderstood, especially their dependence on upstream tension and strain as baseline for draw zone tension.
- Many people believe successive draw zones require stepped increases in speed to maintain tension. This is true for viscoelastic materials, such as cookie dough, but not for most paper, film, and foil webs.
- Draw control is dependent on relative surface speeds of input and output rollers, making them sensitive to roller wear. If speed ratios are set too high, microslip will occur.
- Draw control has a time constant equal to the time it takes the web to pass through the draw zone (Time Constant = Length/Speed). When a change occurs upstream or between the draw zone's driven rollers, the system will require 3x this time constant to reach a new steady state.
- Draw zone don't provide tension feedback; therefore, it can be difficult to tell if the system is working properly.
- In many cases, the draw ratio of roller driven off a common belt or chain is set by the equipment supplier, but undocumented and unknown.

More on this topic was covered in this archived Web Lines column:

Drawing Conclusions: Part 1,Part 2 May-June 2005

Question 6: Rollers

What is a reasonable specification for roller alignment in mm/m, milli-radians, or mils/in?

Based on my experience in 0.5 mil (12 micron) polyester films and many converting processes operating at a 60-in (1.5m) width, I usually recommend 2 mil/ft or about 0.2 milli-radians or 0.2 mm/m (usually measured at the roller shaft or bearing mount width).

I understand that lower numbers are used on wide paper industry equipment. For webs with lower elastic modulus that run at higher elongations, I can imagine loosening this specification to 0.5-1.0 milli-radians.

There are certainly many rollers in any process that are much less sensitive than this specification, but it is difficult to predict in the design phase, which these alignment insensitive rollers will be. Alignment in the plane of the entering and exiting spans are most important (e.g. for a vertical entry and exit of a 180-degree wrap roller, the level is more important than tram to centerline).

More on this topic was covered in this archived Web Lines column:

How Much Misalignment Is Trouble?: Part 1, Part 2 Jan-Feb 2009

Question 7: Rollers

Which rollers in a process will likely have the largest diameters?

Chilled or heated rollers will often be large diameter to have more time in contact and more heat transfer capacity.

Nipped rollers, except those with less than 1 lb/in (175 N/m) load, should have larger diameter (or a larger diameter backup roller) to avoid deflection. This includes coating, laminating, embossing, winding nips and more.

Diameter will go up with roller width and web tensions.

(I haven't written a Web Lines column on this topic, though I have a conference presentation on this from AWEB 2006.)

Question 8: Traction

When is a nipped roller needed on a driven roller separating two tension zones?

The capacity of an unnipped drive roller is an exponential function of web-to-roller coefficient of friction and wrap angle (in radians). For example, a COF of 0.3 and wrap angle of 180-degree (3.14 radians) has the capacity to separate a 2.56 to 1 tension change without slipping.

Therefore, if the tension change between two zones is less than 2.56:1 (plus a safety factor), a nip should not be required.

Care should be taken to ensure the tension change across the unnipped roller not greater than anticipated. Tension measurements taken several rollers away from a driven roller may not represent tension changes due to roller drag or inertia. A nip might be required for machine threading when tension in a zone may be near or at zero, not set point.

Also, care should be taken to ensure that coefficient of friction measured in low speed testing is not lost at high speed and low tension due to air lubrication.

More on this topic was covered in this archived Web Lines column:

Get a Grip: Driving Your Web Jan. 2008

Question 9: Nipped Rollers

What is a simple measurement to determine if two nip rollers are pressing together uniformly?

Footprint. Measure the machine direction contact length vs. cross-roller position. There are many options to measure footprint. One of the simplest is to lay down a piece of colored sticky tape with bubbles intentionally left trapped in the adhesive, then close the nip, open the nip, and measure the machine direction length where the bubbles have been pressed out. Alternately, stick a series of Post-It™ notes in pairs across from each other on eitehr side of a closed nip, tacking them to one of the nipped roller, then open the nip and measure the space between the note where the nip footprint starts and stops.

Gap. For extreme nip variations, shine a flashlight on one side and see if light appears on the other side. Alternately, use a thin feeler gauge or paper strip and see if there is a measureable gap anywhere across the nip footprint.

More on this topic was covered in this archived Web Lines column:

Mission: Detect Nip Variations Sep. 2010

Question 10: Laminating

If a two-layer laminate product curls to the top layer in the machine direction, which would reduce the curl: increase or decrease the top layer pre-laminate tension?

If a laminate curls when tension is removed, it will curl to the product that has the greater web strain. To reduce the curl, reduce the tension in the layer that is on the inside of the curl. Therefore, the answer is to decrease the top layer tension.

More on this topic was covered in this archived Web Lines column:

Get Out of the Scroll Business Mar. 2005

Question 11: Guiding

For automatic guiding in the middle of a process, which is preferred – A steering type guide or a displacement type guide?

A displacement guide is always preferred unless you need to guide a long span. A displacement guide is more compact and has a far lower chance to cause problems through poor setup, poor traction, of wrinkle sensitive webs.

More on this topic was covered in this archived Web Lines column:

Displacement vs. Steering: Battle of the Web Guides **June 2011**

Question 12: Web Tracking

In a vertical span, which will cause the web to track off center more, a roller tram or level error?

Webs track the most from misalignment in the plane of the entering web; therefore, in a vertical span, the level error will cause more tracking than a tram error.

More on this topic was covered in this archived Web Lines column:
Going with the Parallel Flow Aug. 2003

Question 13: Web Tracking

Name four mechanisms that will cause the web to track off centerline through a series of rollers?

Misaligned rollers, asymmetrical diameter variations, asymmetrical bagginess, asymmetrical nip loading of rollers or wound rolls.

More on this topic was covered in this archived Web Lines column:

Your Guide to Web Guiding Part II - Lateral Motion Causes Apr. 2010

Question 14: Wrinkling

If you see wrinkles at low speeds, but the wrinkles go away at higher speeds, what is the most likely reason?

Many wrinkle mechanisms are dependent on traction. As speed increases, air lubrication can reduce web-to-roller traction. Without traction there may not be enough force applied to the web from the roller to cause lateral bending (the source of shear wrinkles) or compress the web laterally beyond the buckling criteria.

More on this topic was covered in this archived Web Lines column:

Optimizing Traction: Goldilocks and the Three Bears Deal with Traction Problems Aug. 2002

Question 15: Wrinkling

If a roller is deflecting into a bowed shape due to gravity, which direction is it best NOT to approach that roller if you don't want to wrinkle: vertically from above, vertically from below, or horizontally?

A bowed roller will tend to gather and wrinkle the web if the bow points towards the entering span. Therefore, a web approaching a roller vertically from below that is deflecting due to gravity is NOT the best approach.

More on this topic was covered in this archived Web Lines column:

Q&A on Roller Deflection Nov. 2010

Question 16: Spreading

Name a spreader or anti-wrinkle roller that does NOT have a rubber surface?

Every roller has some anti-wrinkle benefit due to inducing machine direction curvature in the web, providing shape stiffening.

A concave roller can be machined out of any material, including aluminum or steel.

A hyper-crowned roller, where the diameter variations are several times greater than web strain, can induce spreading, similar to a D-shaped bar pushed into the web.

There are some segmented bowed rollers with a series of cylindrical metal segments mounted, shish kabob style, on a curved shaft, though most bowed rollers have rubber sleeves.

In some cases, a metal roller with raised chevron ridges will prevent wrinkles by allowing the web to pass over a roller with many minor non-damaging buckles instead of a major crease or break causing wrinkle.

Chevron ridge rollers pointing upstream can have a plowing and spreading effect if they are not turning (then technically not a roller), driven below web speed, or reverse driven.

More on this topic was covered in this archived Web Lines column: Concave Rollers Pros & Cons July 2005

Question 17: Winding

If you center wind at constant torque, what is the percent change in tension if the roll diameter changes from 100mm to 400mm?

Constant torque center winding applies a constant torque in units of force times distance (e.g. lbs-in, N-m) to the center of the winding roll. This torque is converted into tension based on the distance, which is the radius, to the outside surface of the winding roll. The actual radial value is as important to this question as the ratio of the radial change – 1:4. If the torque is value is initially divided by 1 to get a tension and later divided by 4 to create tension, the final tension will be 4:1 or four times lower.

If the starting torque was 400 N-mm, the initial tension would be (400 N-mm)/(100mm) = 4 N. At the final radius, tension would be (400 N-m)/(400mm) = 1 N; so the tension would drop 4x.

(I need to write a column on tension, torques, and tapers.)

Question 18: Winding

If you wind a paper and a film product of equal thickness and roll geometry at the same combination of winding conditions (tension, nip, taper, speed), which will likely to have higher internal roll pressures or greater tightness?

The mostly likely answer is the film product will wind a much tighter roll, especially if it is a lower modulus film, such as polyethylene or polypropylene.

Paper products tend to wind looser rolls since they have a high relative compressibility in the radial direction and the web stretch (a.k.a. strain) in the paper layers can lose their tension from radial shifting or compressing of the roll's core and inner layers.

Film products tend to wind tighter rolls since they have a low relative compressibility in the radial direction and the web strain (especially in stretchy films) in the film will have minimal tension loss from radial shifting or compressing of the roll's core and inner layers.

One exception might be high speed winding of a smoother film without control of the entrained air layer compared to a more porous paper. Where paper can absorb an entrained air layer, films will capture the air between layers. Over time, the air can bleed out of a film roll, loosening it, potential to a looser condition than a paper on the same winder.

More on this topic was covered in this archived Web Lines column:

The Pressure of Winding Rolls May 2006

Question 19: Winding

Name three advantages of using a nipped or gap controlled roller ahead of winding?

- Winding nip and gap rollers at winding will reduce lateral shifting of layer from air escape and winding roll diameter variations.

- Winding nip and gap rollers will reduce wrinkles from winding roll diameter variations.
- Winding gap rollers provide these first two benefits without the high footprint pressure than can damage some materials.
- Winding gap rollers can reduce interlayer slip in the top layer of a winding roller, discouraging the slip ‘knot’ buckling defects.
- Winding nip rollers will reduce entrained air, especially in low porosity product winding, and create a roll that changes less over time.
- Winding nip rollers can increase roll tightness without addition torque applied at the core.
- Winding nip rollers can press down on high diameter bands, creating a more cylindrical wound roll.

More on this topic was covered in this archived Web Lines column:

Winding Better Rolls June 2003

Question 20: Unwinding

What is an easy way to determine if an unwinding roll is cinching (i.e. some of the roll’s layers are slipping in the machine direction?

Draw a radial line (often called a spoke line) on the side of the roll before applying unwind tension. If the spoke line remains straight and perpendicular to the tangent of the core, then the layers are not slipping in the machine direction. If the line forms a curve or step at any point, that indicates the layers are slipping within the roll. Machined direction slipping often enables lateral shifting; therefore, telescoping, especially near the core, is a likely indicator or cinching.

More on this topic was covered in this archived Web Lines column:

Cinching Belt Tightening Gone Bad: Part 1, Part 2 Feb.-Mar. 2003